

Deterministic – computed once, cached (no leakage possible)

F1 raw & log
10 raw + log1p

F2 inter-arrival
gaps · stats · fractions

F3 length distribution
9 stats · entropy · equality

F4 sequence shape
10 diffs · 10 ratios · trends

F5 protocol semantics
bands · mod 4/8/16 · RTP-12

F6 burst structure
frame groups at 0.5/1/5/20 ms

F7 rate · throughput
bytes/s · inst rate l_i/d_i

F8 spectral
DCT · FFT of 5-seq

241 columns

Shipped feature set = F1-F8

Fitted – inside the CV loop, out-of-fold

F9 class-conditional
Markov likelihoods

F10 class-conditional
Gaussian mixtures

F11 neighbourhood
per-class k-NN distances

Ablation: -1.8 pt accuracy → disabled

Auxiliary-derived – leak-safe probes

F12 frozen-encoder
linear probes (46 cols)

Ablation: -0.01 to 0.00 → retired